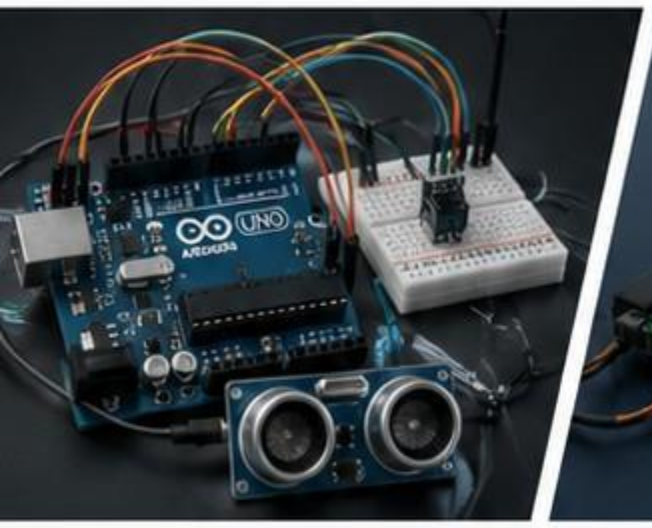
NED UNIVERSITY
OF ENGINEERING & TECHNOLOGY

Department of CIS Engineering

Course:
Microprocessor Based System Design (CS-301)

Project:

"Smart Helmet System Using Arduino and RF Wireless Communication"



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Abstract

Road accidents caused by drunk driving and lack of attention during driving are increasing day by day. Motorcyclists are especially at high risk because they are more exposed to injuries during accidents. The purpose of this project is to design and implement a Smart Helmet System that improves road safety by detecting alcohol consumption and nearby obstacles.

This project uses Arduino Uno microcontrollers, RF transmitter and receiver modules, an alcohol sensor, an IR obstacle sensor, LCD display, buzzer, and motor driver module. The transmitter side is mounted on the helmet and continuously monitors the rider's condition. If alcohol is detected or an obstacle appears near the helmet, the information is wirelessly transmitted to the receiver side through RF communication.

The receiver side receives the data and displays warning messages on the LCD screen. A buzzer is activated when alcohol is detected, and the motor is stopped when an obstacle is detected. The system therefore helps in reducing accidents and improving rider safety.

The project demonstrates the practical implementation of embedded systems, wireless communication, sensor interfacing, and real-time monitoring using Arduino.

Introduction

Technology is playing an important role in improving transportation safety systems. Many accidents occur because riders drive under the influence of alcohol or fail to notice nearby obstacles. Traditional helmets only provide physical protection after an accident, but modern smart helmets can help prevent accidents before they happen.

The Smart Helmet System is designed to enhance rider safety by integrating multiple sensors and wireless communication techniques. The system uses an alcohol sensor to determine whether the rider has consumed alcohol and an IR sensor to detect nearby obstacles. The transmitter module sends this information wirelessly to the receiver module.

At the receiver side, the system processes the received data and performs appropriate actions such as:

- Activating a buzzer when alcohol is detected
- Displaying warning messages on an LCD
- Turning the motor OFF when obstacles are detected

This project combines hardware and software concepts including Arduino programming, sensor interfacing, RF communication, and motor control.

Objectives

The main objectives of the Smart Helmet System are:

1. To detect alcohol consumption using an alcohol sensor.
2. To detect obstacles using an IR sensor.
3. To establish wireless communication using RF transmitter and receiver modules.
4. To display system status and warning messages on an LCD.
5. To activate a buzzer when alcohol is detected.
6. To stop the motor when obstacles are detected.

7. To improve rider safety using embedded systems technology.
8. To implement a real-time monitoring system using Arduino.

Problem Statement

Motorcycle accidents often occur due to careless driving, alcohol consumption, and failure to detect obstacles in time. Existing helmets only protect riders after accidents but do not actively help in accident prevention.

The problem addressed in this project is:

"How can a helmet system intelligently detect unsafe driving conditions and provide real-time safety measures using wireless communication and sensors?"

The Smart Helmet System solves this problem by continuously monitoring alcohol levels and nearby obstacles and sending warnings to the rider through wireless communication.

System Overview

The Smart Helmet System consists of two major sections:

1. Transmitter Section

The transmitter side is mounted on the helmet and includes:

- Arduino Uno
- Alcohol sensor
- IR obstacle sensor
- RF transmitter module

The transmitter continuously reads sensor values and sends wireless data to the receiver module.

2. Receiver Section

The receiver side includes:

- Arduino Uno
- RF receiver module
- LCD display
- Buzzer
- Motor driver module
- DC motor

The receiver processes incoming data and performs actions such as displaying messages, activating warnings, and controlling the motor.

Components Used

Component	Purpose
Arduino Uno	Main microcontroller for processing
Alcohol Sensor	Detects alcohol presence
IR Obstacle Sensor	Detects nearby obstacles
RF Transmitter Module	Sends wireless data
RF Receiver Module	Receives wireless data
LCD 16x2 with I2C	Displays system messages
Buzzer	Provides warning sound
L298N Motor Driver	Controls motor
DC Motor	Simulates bike engine
Breadboard	Hardware connections
Jumper Wires	Electrical connections
Battery Pack	Power supply
Antenna Wire	Improves RF range

Hardware Description

Arduino Uno

Arduino Uno is the main controller used in this project. It reads sensor data, processes conditions, and controls outputs. It uses the ATmega328P microcontroller.

Features:

- 14 digital I/O pins
- 6 analog input pins
- USB programming support
- Easy sensor interfacing

Alcohol Sensor

The alcohol sensor detects alcohol concentration in the environment. If the alcohol level exceeds the threshold value, the system identifies unsafe conditions.

Working Principle:

- Sensor output increases when alcohol gas is present.
- Arduino reads the analog value.
- If the value exceeds the threshold, alcohol is considered detected.

IR Obstacle Sensor

The IR sensor detects nearby obstacles using infrared light.

Working Principle:

- IR transmitter emits infrared rays.
- IR receiver detects reflected rays.
- If reflection occurs, obstacle is detected.

RF Transmitter and Receiver Modules

These modules provide wireless communication between the helmet and receiver section.

Working Principle:

- RF transmitter converts digital data into radio frequency signals.
- RF receiver receives the signals and converts them back into digital data.

Operating Frequency:

- 433 MHz

LCD Display

A 16x2 LCD with I2C module is used to display system messages.

Displayed Messages:

- System Safe
- Alcohol Detected
- Obstacle Found
- Motor Running
- Motor Stopped

Buzzer

The buzzer provides an audible warning when alcohol is detected.

Motor Driver and DC Motor

The motor driver controls the DC motor based on obstacle detection.

- Motor ON → Safe condition
- Motor OFF → Obstacle detected

Circuit Explanation:

Receiver Side Connections:

RF RX Pin	Arduino Pin
DATA	D11
VCC	5V
GND	GND
LCD Connections	
LCD Pin	**Arduino Pin**
SDA	A4
SCL	A5
VCC	5V
GND	GND
Buzzer	
Buzzer Pin	**Arduino Pin**
Positive	D8
Negative	GND
Motor Driver	
Motor Driver Pin	**Arduino Pin**
IN1	D7
IN2	D6

Transmitter Side Connections:

Alcohol Sensor	
Alcohol Sensor Pin	**Arduino Pin**
AO	A0
VCC	5V
GND	GND
IR Sensor	
IR Sensor Pin	**Arduino Pin**
OUT	A1
VCC	5V
GND	GND
RF Transmitter	
RF TX Pin	**Arduino Pin**
DATA	D12
VCC	5V
GND	GND

Working Principle

Transmitter Code Explanation

The transmitter code performs the following tasks:

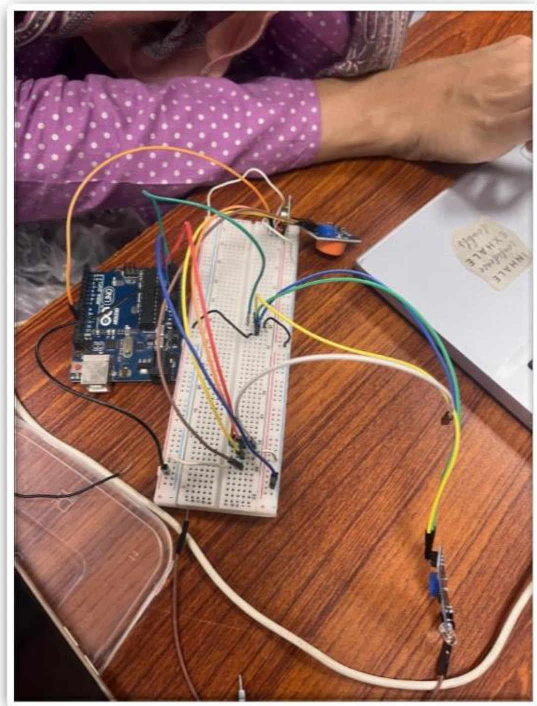


Fig 1: Transmitter of the Project

1. Include Libraries

The following libraries are included:

- RH_ASK library for RF communication
- SPI library

2. Sensor Initialization

The sensor pins are defined:

- Alcohol sensor → A0
- IR sensor → A1

3. RF Initialization

The RF module is initialized for wireless transmission.

4. Sensor Reading

Arduino continuously reads:

- Alcohol sensor values
- IR sensor values

5. Condition Checking

The code checks:

- Whether alcohol is detected
- Whether an obstacle is detected

6. Data Transmission

The information is converted into a string format and transmitted wirelessly.

Example:

1,1

Meaning:

- Alcohol detected
- Obstacle detected

Receiver Code Explanation

The receiver code performs the following tasks:

1. Initialize LCD and RF Module

The LCD and RF receiver are initialized.

2. Receive Data

The receiver continuously waits for RF data.

3. Decode Data

The received message is split into:

- Alcohol status
- Obstacle status

4. Perform Actions

- **If Alcohol Detected**
  - Buzzer ON
  - LCD warning message
- **If Obstacle Detected**
  - Motor OFF
  - LCD obstacle message
- **If Safe**
  - Motor ON
  - LCD safe message

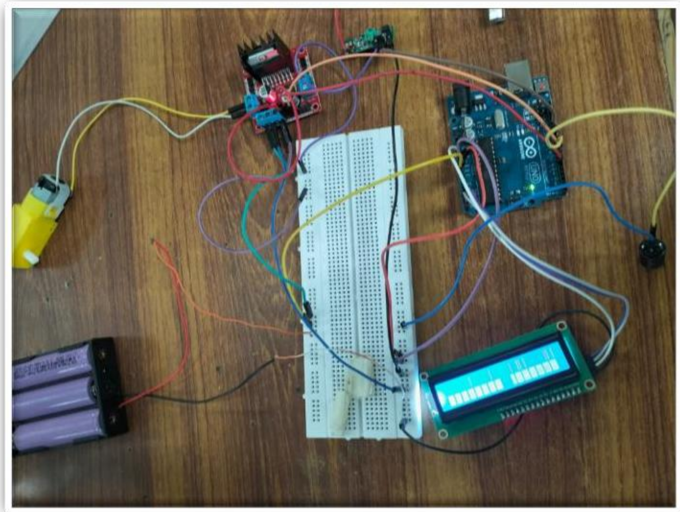
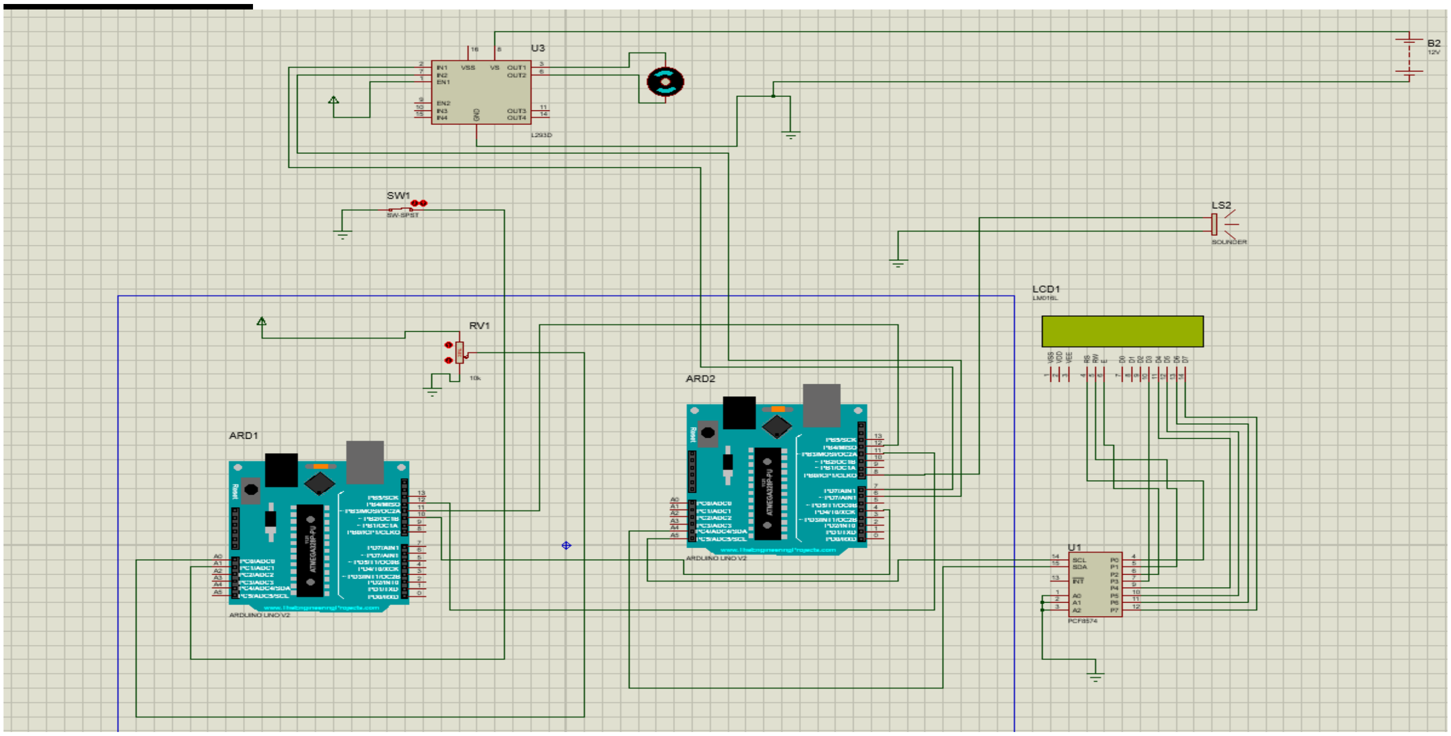


Fig 2: Receiver of the Project

Simulation



Software Used

Software	Purpose
Arduino IDE	Programming and uploading code
Proteus	Circuit simulation
Serial Monitor	Debugging and monitoring

Advantages

1. Improves rider safety.
2. Provides real-time monitoring.
3. Prevents drunk driving.
4. Detects nearby obstacles.
5. Wireless communication reduces wiring complexity.
6. Low-cost implementation.
7. Easy to expand with additional sensors.

Limitations:

1. RF communication range is limited.
2. Sensors may be affected by environmental conditions.
3. Battery backup is limited.
4. Wireless interference may affect communication.

Applications:

1. Motorcycle safety systems
2. Smart transportation systems
3. Accident prevention systems
4. Industrial safety helmets
5. Wireless monitoring systems

Future Scope:

The Smart Helmet System can be further improved by adding advanced technologies.

Possible future enhancements include:

1. GPS tracking for accident location.
2. GSM module for emergency SMS alerts.
3. Bluetooth mobile application integration.
4. Camera-based monitoring system.
5. Cloud-based data storage.
6. IoT integration for remote monitoring.
7. Advanced AI-based obstacle detection.
8. Voice alert system.
9. Automatic accident detection.
10. Rechargeable smart battery system.

Conclusion:

The Smart Helmet System successfully demonstrates the integration of embedded systems, sensors, and wireless communication to improve rider safety.

The project detects alcohol consumption and nearby obstacles using sensors and transmits the information wirelessly through RF modules. The receiver side processes the information and generates appropriate responses such as activating the buzzer, displaying warnings on the LCD, and controlling the motor.

This project provides a practical and cost-effective approach for accident prevention and demonstrates the importance of smart safety systems in modern transportation.

The implementation of this system proves that technology can significantly contribute to safer driving conditions and reduced accident risks.